Unsupervised Spectral–Spatial Semantic Feature Learning for Hyperspectral Image Classification

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***Abstract*— Can we automatically learn meaningful semantic feature representations when training labels are absent? Several recent unsupervised deep learning approaches have attempted to tackle this problem by solving the data reconstruction task. However, these methods can easily latch on low-level features. To solve this problem, we propose an end-to-end spectral–spatial semantic feature learning network (S3FN) for unsupervised deep semantic feature extraction (FE) from hyperspectral images (HSIs). Our main idea is to learn spectral-spatial features from high-level semantic perspective. First, we utilize the feature transformation to obtain two feature descriptions of the same source data from different views. Then, we propose the spectral–spatial feature learning network to project the two feature descriptions into the deep embedding space. Subsequently, a contrastive loss function is introduced to align the two projected features, which should have the same implied semantic meaning. The proposed S3FN learns the spectral and spatial features separately, and then merges them. Finally, the learned spectral–spatial features by S3FN are processed by a classifier to evaluate their effectiveness. Experimental results on three publicly available HSI datasets show that our proposed S3FN can produce promising classification results with a lower time cost than other state-of-the-art (SOTA) deep learning-based unsupervised FE methods.**

***Index Terms*— Deep learning, high-level semantic, hyperspectral image (HSI) classification, unsupervised feature learning.**

I. Introduction

With the rapid development of hyperspectral remote-sensing technology, both spectral and spatial information of the ground objects is now accessible. Due to the high spectral resolution of hyperspectral images (HSIs), objects can be identified in the HSIs more easily than in RGB images, thus launching a new era in remote-sensing applications [1], such as environmental monitoring, precision agriculture, and urban mapping [2]–[6]. HSI classification, which assigns labels to all pixels, has become one of the most important research directions in this domain. Compared with RGB classification, HSI classification is more challenging due to spectral variations caused by changes in light, environment, and atmospheric conditions and sensor noise [7]. Moreover, the curse of dimensionality indicates that it is challenging to acquire helpful information from original HSIs [8]. Because of these problems, feature extraction (FE) is proposed as a preprocessing step to improve the separability of various categories in classification tasks, which is one of the most active research areas in HSI processing [9].

Numerous supervised FE methods have been presented in the literature [10], [11], including local Fisher discriminant analysis (LFDA) [12], neighborhood preserving embedding (NPE) [13], spatial and spectral regularized local discriminant embedding (SSRLDE) [14], sparse and low-rank graph-based discriminant analysis (SLGDA) [15], and maximum likelihood estimation-based joint sparse representation (MLEJSR) [16]. These all can be regarded as shallow models trained with labels [3], [17]. Although such well-designed hand-crafted FE methods have the ability to represent geometric and textural attributes of ground objects, they also have some drawbacks. First, due to the heavy reliance on prior information from the experts, these methods cannot balance between discriminability and robustness. Second, these FE methods are designed with a shallow monolayer structure, and therefore less robust than deep models [18], [19]. Deep learning models that learn effective features from large-scale datasets have been developed. Classical deep learning methods include convolutional neural networks (CNNs) [20], [21], recurrent neural networks (RNNs) [22], and fully convolutional networks (FCNs) [23]. Recent research has demonstrated the potential of deep models for representation in the remote-sensing field [24]–[31]. Compared with shallow models, deep models enable to automatically extracting deep abstract features from images, thus increasing the accuracy of subsequent classification tasks [17].

What happens however, when there is no access to large numbers of accurate semantic labels at training time? Unsupervised FE techniques therefore have recently emerged as an alternative to supervised FE, as it is often impossible to obtain enough labeled training samples to improve the accuracy of supervised FE methods when dealing with HSI in practice.

Unsupervised FE research has also evolved from shallow to deep. The shallow unsupervised FE methods like principal component analysis (PCA) [32], independent component analysis (ICA) [33], locally linear embedding (LLE) [34], superpixelwise PCA (SuperPCA) [35], local neighborhood structure preserving embedding (LNSPE) [36], etc., learn shallow features but rely on the domain knowledge of experts, which limits their applicability. In contrast, unsupervised FE models based on deep learning show potential as they are able to automatically learn deep features from the data itself.

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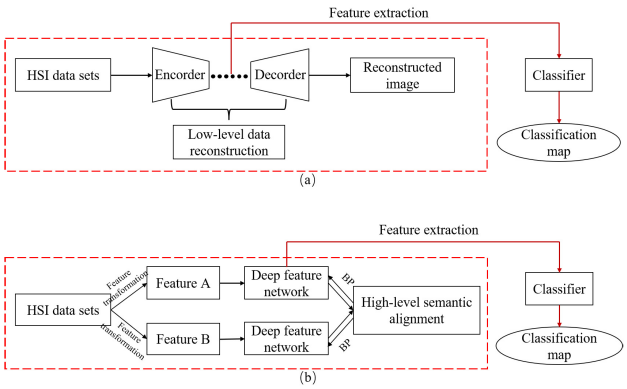


Fig. 1. Low-level data reconstruction in (a) learns a representation of input data by reconstructing input patterns with a minimum structure error, which easily latches on low-level features. The proposed high-level semantic alignment in (b) considers the semantic correlation of representations of the same pixels from different views and aligns representations in the feature space to learn semantic features.

Romero *et al.* [37] proposed an unsupervised convolutional network for learning spectral–spatial features using a greedy layer-wise technique to update the parameters of the network. To capture the interaction information neglected by the greedy layer-wise unsupervised learning, Yu *et al.* [38] proposed an unsupervised convolutional feature fusion network (UCFFN) to integrate the features from the outputs of different layers. In [39], the residual Conv-Deconv network (RCDN) was proposed and trained in an end-to-end manner to improve the training efficiency. In [40], a recursive autoencoder (RAE) network was proposed and a local homogeneous area of the image was used to learn features. Zhang *et al.* [41] proposed a modified generative adversarial network (GAN) to train a deep learning-based feature extractor without supervision. Spatial information is lost during flattening operations in stacked autoencoders (SAEs) [42]. Fully connected layers, however, can be replaced with 3-D operations, and 3-D convolutional autoencoding (3-D-CAE) can be deployed to extract hierarchical spectral and spatial features [43]. Based on 3-D-CAE, Zhang *et al.* [44] further developed the 3-D symmetric all convolution neural network (3-D-SACNN) for deep learning-based unsupervised feature learning.

The task of learning effective unsupervised spectral–spatial features remains a significant and open challenge in HSI classification. In general, typical deep learning-based unsupervised FE methods use an encoder–decoder paradigm to generate intermediate feature representations solely from the original images [37], [38], [40], [43], [44], as shown in Fig. 1(a). These FE methods rely on point-to-point data reconstruction, and are extremely sensitive to outliers [6], [45], [46]. Furthermore, they are likely to capture low-level features such as color, contrast, and texture, etc., as shown in Fig. 2(b). Unfortunately, these low-level features are insufficient for the high-level semantic classification. Although 3-D operations used in most of these methods can extract spectral–spatial information simultaneously, they also add cumbersome tunable parameters to the network, thus reducing efficiency [47].

To solve these problems, we propose the spectral–spatial semantic feature learning network (S3FN) for learning semantic features. In contrast to current FE methods based on low-level image reconstruction tasks, we focus on high-level semantic information for unsupervised FE in HSI classification. As shown in Fig. 1(b), we first obtain the feature descriptions of the same source data under multiviews via feature transformation. Second, we propose the learnable spectral–spatial feature learning network to project the two feature descriptions into the deep embedding space. Subsequently, a contrastive loss function [48], [49] is introduced to align the two projected features in the deep embedding space, which should be regarded as the same implied semantic meaning. In this case, the learning loss is constructed without labels. Furthermore, we designed a feature learning network to learn spectral and spatial features separately to improve the efficiency of our proposed FE method. The trained feature learning network is regarded as a feature extractor, and the learned spectral–spatial features are validated by a classifier such as support vector machine (SVM) [50]. To further analyze the features, we visualize the feature maps in Fig. 2. It can be observed that the high-level features extracted by our proposed spatial semantic feature learning network (SpaSFN) in Fig. 2(c) are more abstract to a portion of the ground objects in the original image, whereas the shallow features extracted by 3-D-CAE in Fig. 2(b) contain detailed boundary information. The proposed semantic feature learning framework reconstructs semantic features instead of the original data. Therefore, our framework can learn high-level semantic features, and boost the final classification results using unsupervised learned features. The major contributions can be summarized as follows.

1) Compared with low-level data reconstruction-based FE methods, the proposed S3FN framework can learn more discriminative high-level spectral–spatial features with implied semantic meaning, thus are more suitable for downstream semantic classification task.
2) An efficient spectral–spatial feature learning network was designed to learn the hierarchical spectral and spatial features.
3) Experimental results show that S3FN yields competitive results on various datasets with a lower time cost than previous deep learning-based unsupervised FE methods.

The rest of this article is organized as follows. In Section II, we briefly introduce the related work on unsupervised deep learning-based FE for HSI classification tasks. The detailed descriptions of the proposed spectral semantic feature learning, spatial semantic feature learning, and spectral–spatial semantic feature learning frameworks are given in Section III. The related experiment results and discussion are provided in Section IV. Final conclusions are drawn in Section V.

II. Related Work

A. Data Reconstruction-Based FE Methods

In the remote-sensing filed, researchers have tried to accomplish unsupervised feature learning with deep neural networks (DNNs) for HSI classification [40], [51], [52]. An autoencoder (AE) artificial neural network permits efficient learning and encoding of data in an unsupervised manner [53].

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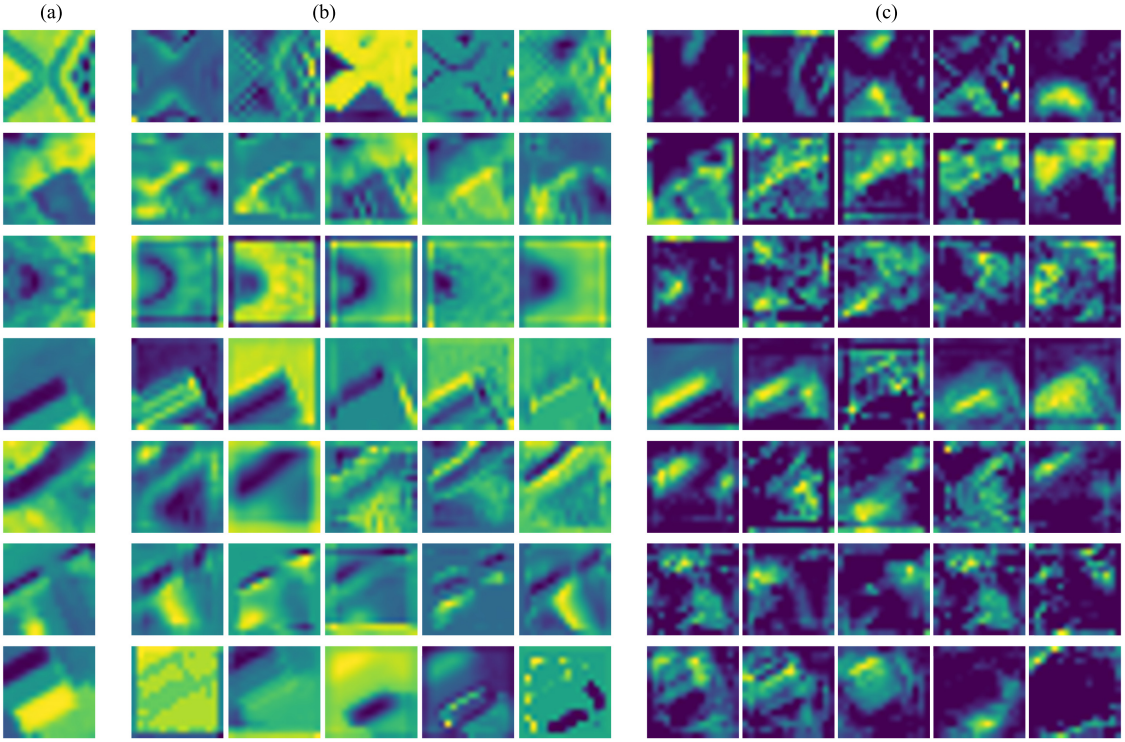


Fig. 2. (a) Input patches from the Pavia University dataset. (b) Learned low-level representations via the reconstruction-based FE method 3-D-CAE [43]. (c) Learned high-level representations via the proposed Spa-SFN.

It learns a representation of input pixels by reconstructing their input patterns with a minimum of reconstruction error. As shown in Fig. 1(a), the architecture of reconstruction-based methods includes an encoder with a companion decoder. The encoder projects inputs into a feature space to acquire low-dimensional representations, and the decoder restores the input. This strategy for the network architecture eliminates the need for labeled samples.

To train the reconstruction network, mean square error (MSE) is usually chosen as the loss function with L_2 regularization to achieve sparsity. However, only the point-to-point error is considered in the MSE loss function during training, resulting in sensitivity to outliers. Since the task is to reconstruct the data, the network is prone to capture low-level features such as color and texture, which is inefficient for high-level semantic classification. To solve these problems, we assume that the features of the same pixel from different views have the same high-level semantic meaning, and therefore propose to learn features from a high-level semantic perspective.

B. 3-D Operations-Based FE

To extract hierarchical spectral–spatial features, current FE methods adopt 3-D operations to convolve the neighborhood of pixels in both spectral and spatial dimensions. In the reconstruction network, 3-D deconvolution is usually combined with 3-D convolution. Examples of 3-D convolution and 3-D deconvolution are shown in Fig. 3. In Fig. 3(a), given a $5 \times 5 \times 4$ input, a $3 \times 3 \times 3$ kernel slides on it with stride 1 to obtain a output with $3 \times 3 \times 2$ size. As we see in Fig. 3(b), the

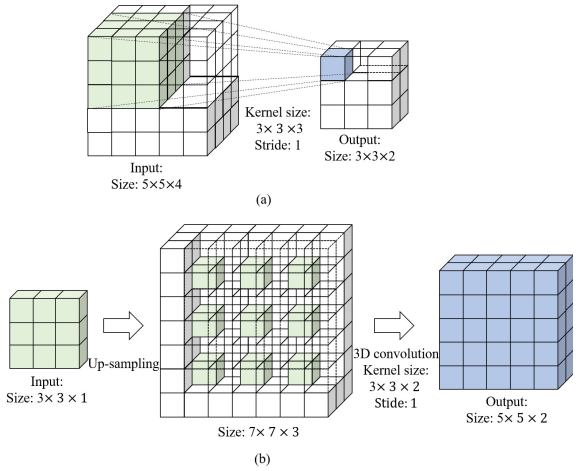


Fig. 3. Illustration of 3-D convolution in (a) and deconvolution in (b). Deconvolution includes 3-D upsampling and 3-D convolution processes.

deconvolution includes upsampling and convolution processes. The 3×3 input is first upsampled to a larger size such as 7×7 with a zero filling operation. After then, convolution is performed on the upsampled input to obtain a 5×5 sized output. To extract hierarchic features from the input, encoder was designed as a deep convolution structure with multiple 3-D convolutions. Moreover, the companion decoder includes multiple 3-D deconvolutions to reconstruct the input.

Although 3-D operations can extract hierarchical spectral–spatial features simultaneously, they add cumbersome tunable parameters to networks. The extensive redundancy in the adjacent bands makes direct convolution on the raw 3-D

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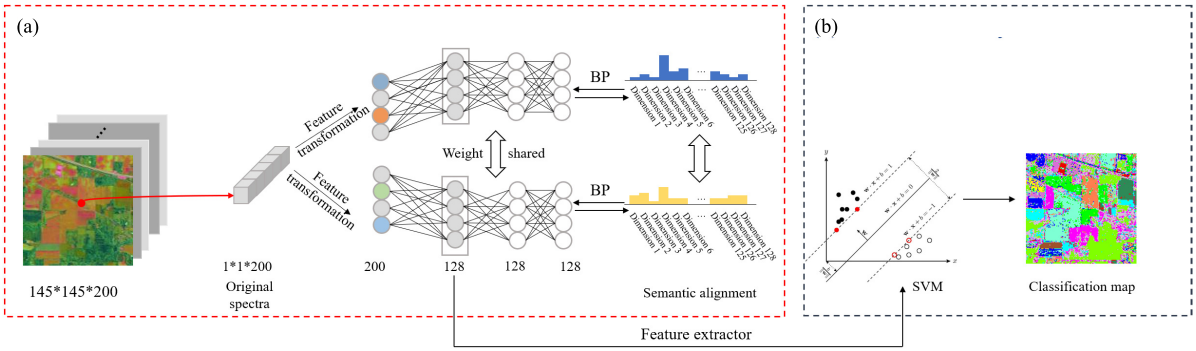


Fig. 4. (a) Framework of the proposed Spe-SFN. A random spectral feature transformation is applied to obtain different feature descriptions of the original spectral signal, which are separately processed by the spectral feature learning network. Subsequently, the semantic alignment strategy is proposed to align the elements of the same semantic category more closely in the embedding feature space. (b) After training, the extracted spectral features are processed by a classifier to yield the classification map.

HSIs inefficient and unnecessary. To improve the efficiency, we designed an efficient spectral–spatial feature learning network to extract spectral and spatial features separately.

III. Methodology

A. Unsupervised Semantic Feature Learning Framework

To efficiently learn semantic features from an unlabeled dataset, we propose an unsupervised feature learning framework as shown in Fig. 1(b). Each image pixel $x \in R^n$ is considered as a pattern, and its spectrum with n bands is regarded as the initial feature set, also known as the spectral signal. First, we apply the feature transformation to obtain two feature descriptions of x from different views. Second, our proposed feature learning network projects these feature descriptions into a deep embedding space. Feature representations of the same pixel from different views should have the same high-level semantics meaning. Thus, a contrastive loss function is introduced to align the two projected features. After these three steps, the proposed framework can exploit the inherent representations of pixels under random feature transformations, which can be applied to subsequent classification tasks.

The learned deep features are processed by a classifier to verify the effectiveness of the results from the unsupervised feature learning framework. Under the proposed framework shown in Fig. 1(b), Section III-B details the proposed spectral semantic feature learning framework for HSIs. The spatial semantic feature learning framework is presented in Section III-C, and the spectral–spatial semantic feature learning framework is discussed in Section III-D.

B. Spectral Semantic Feature Learning for HSI Classification

Previous research has shown that DNNs have great potential in spectral feature learning of HSIs [54]. However, these methods are trained in a supervised fashion, while the ground-truth labels are available during training. In this section we present the proposed unsupervised spectral semantic feature learning network (Spe-SFN) for extracting the semantic spectral features in HSIs.

For each pixel $x \in R^n$, a feature transformation is first applied to obtain different views of feature descriptions, which can be formulated as

$$v_{\text{spe}} = \mathcal{N}_{\text{spe}}(x_{\text{spe}}) \tag{1}$$

$$v^0_{\text{spe}} = \mathcal{N}_{\text{spe}}(x_{\text{spe}}) \tag{2}$$

where x is written as x_{spe} here to distinguish from the notation x_{spa}, and $\mathcal{N}_{\text{spe}}(\cdot)$ represents the random spectral feature transformations realized by adding Gaussian noise with a mean value 0 and a standard deviation 0.5 to each pixel of the original image. Specifically, we apply $\mathcal{N}_{\text{spe}}(\cdot)$ twice to generate two different feature descriptions of the same pixel, i.e., v_{spe} and v^0_{spe}, with (1) and (2).

Subsequently, a simple spectral feature learning network was designed to extract hierarchical deep spectral features. The architecture of the proposed Spe-SFN is shown in Fig. 4. There are two identical branches in the network with shared weights and three fully connected layers in each branch. After a fully connected layer, the rectified linear units (ReLUs) [55] was selected as the activation function to increase the nonlinearity of the neural network, which can be defined as $g_{(x)} = \max(0, x)$. For illustrative purposes, the first connected layer is called ϕ_{spe}, and the last two fully connected layers are called f_{spe}. Thus, the multiview feature descriptions of the same source pixels are projected into a deep embedding space by the feature learning network and the embedding features can be formulated as

$$c_{\text{spe}} = f_{\text{spe}}(\phi_{\text{spe}}(v_{\text{spe}})) \tag{3}$$

$$c^0_{\text{spe}} = f_{\text{spe}}\left(\phi_{\text{spe}}\left(v^0_{\text{spe}}\right)\right). \tag{4}$$

In the supervised feature learning, a cross-entropy loss function is usually utilized to learn the mapping between the data X and the corresponding classes C. In most cases however, the training labels are difficult to obtain. As a solution, we define a semantic prior that features c_{spe} and c^0_{spe} of the same pixel from different views should have the same high-level semantics. Therein, we introduce a contrastive loss function [48], [49] to enforce consistency between deep features of transformed pixels under different views in the embedding space as the following:

$$\min \mathcal{L}_{\text{spe}}\left(f_{\text{spe}}(\phi_{\text{spe}}(v_{\text{spe}})), f_{\text{spe}}\left(\phi_{\text{spe}}\left(v^0_{\text{spe}}\right)\right)\right). \tag{5}$$

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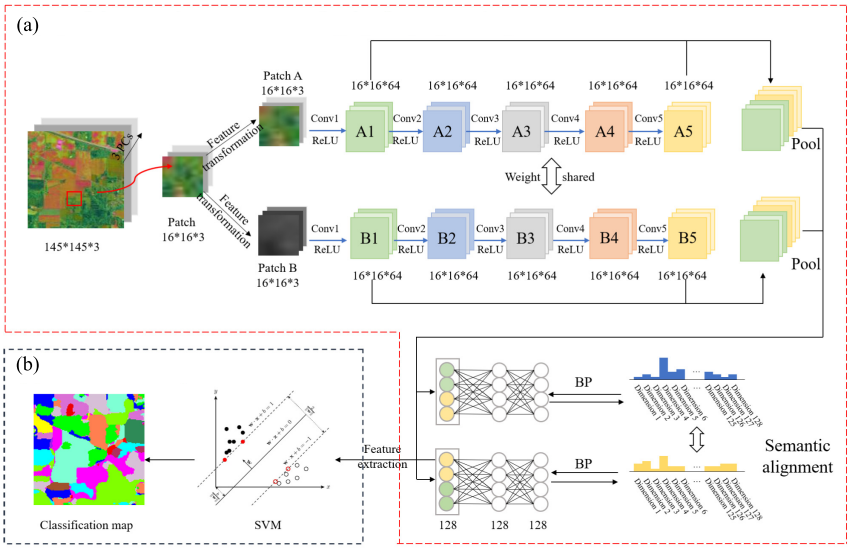


Fig. 5. (a) Framework of the proposed Spa-SFN. The random spatial and intensity feature transformations are applied to obtain different feature descriptions of the original spatial signal, which are separately processed by the spatial feature learning network. Subsequently, the semantic alignment strategy is proposed to align the deep feature of transformed data. (b) After training, the extracted spatial features are processed by a classifier to evaluate the effectiveness of the proposed framework.

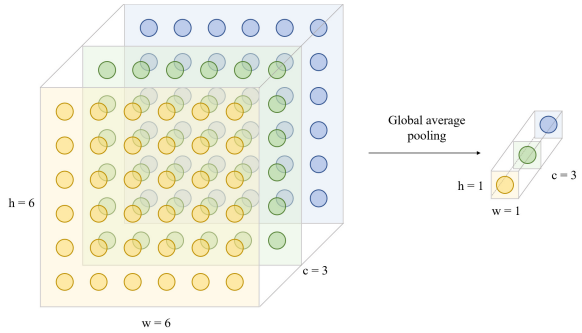


Fig. 6. Illustration of global average pooling.

Based on (5), the loss function can be written as

$$\mathcal{L}_{\text{spe}} = -\sum_{i \in I} log \frac{\exp\left(c_{\text{spe}(i)} \cdot c^0_{\text{spe}(i)} / \tau\right)}{\sum_{a \in Ai} \exp\left(c_{\text{spe}(a)} \cup c^0_{\text{spe}(I)}\right) / \tau)} \tag{6}$$

where $i \in I \equiv \{1, \ldots, N\}$ is the index of an arbitrary pixel within a mini-batch with size N, the $\cdot$ symbol denotes the dot product, $\cup$ represents union operation, τ is a scalar temperature parameter, and $A(i) \equiv I \backslash \{i\}$. After the training, we extract the output of the first fully connected layer as feature representation, which are assumed to retain most of the spectral information. The extracted features are processed by the supervised classifier to yield classification results.

C. Spatial Semantic Feature Learning for HSI Classification

Without spatial information, the spectral heterogeneity and the complex spatial distribution of objects make it difficult to achieve high accuracy in HSI classification. Therefore, the framework must learn the robust and discriminant spatial features to realize optimal classification results. In this section, we describe the proposed Spa-SFN.

The framework of Spa-SFN is shown in Fig. 5. Considering the redundancy in high-dimensional HSIs, the PCA is applied to reduce dimension of the data, and the first p ($p = 3$ in this article) principal components are retained. For each pixel, the neighboring pixels from a window of size $w \times w \times p$ are extracted, where w is the patch size ($w = 16$ in this article). In this way, the array $x_{\text{spa}} \in R^{w \times w \times p}$ can be regarded as the spatial signal of the original pixel.

Feature transformations are also applied to obtain feature descriptions of the same source pixel in different views, which can be formulated as

$$v_{\text{spa}} = \mathcal{N}_{\text{spa}}(x_{\text{spa}}) \tag{7}$$

$$v^0_{\text{spa}} = \mathcal{N}_{\text{spa}}(x_{\text{spa}}) \tag{8}$$

where $\mathcal{N}_{\text{spa}}(\cdot)$ represents the combination of random spatial and intensity transformations. Specifically, the spatial transformations represent random transformations in the spatial domain, such as flips and rotations. The intensity transformations represent random transformations in the intensity domain, such as enhancement or diminution of brightness and contrast. Similarly, we also apply $\mathcal{N}_{\text{spa}}(\cdot)$ twice to generate two different spatial features of the same pixel, i.e., v_{spa} and v^0_{spa}, with (7) and (8).

To learn hierarchical spatial features, the feature learning network was designed with two identical branches, consist of five 2-D convolutional layers, one global average pooling layer, and two fully connected layers with shared weights. In the following, we will describe the network structure in detail.

In the proposed spatial feature learning network, we only utilize the 2-D convolutional kernel in the convolutional layer. The convolutional feature extracted from each convolutional layer can be formulated as

$$u^{xy}_{ij} = g\left(\sum_{m=1}^{M_{i-1}} \sum_{p=0}^{P_i - 1} \sum_{q=0}^{Q_i - 1} w^{pq}_{ijm} \left(v^{(x+p)(y+q)}_{\text{spa}(i-1)m}\right) + b_{ij}\right) \tag{9}$$

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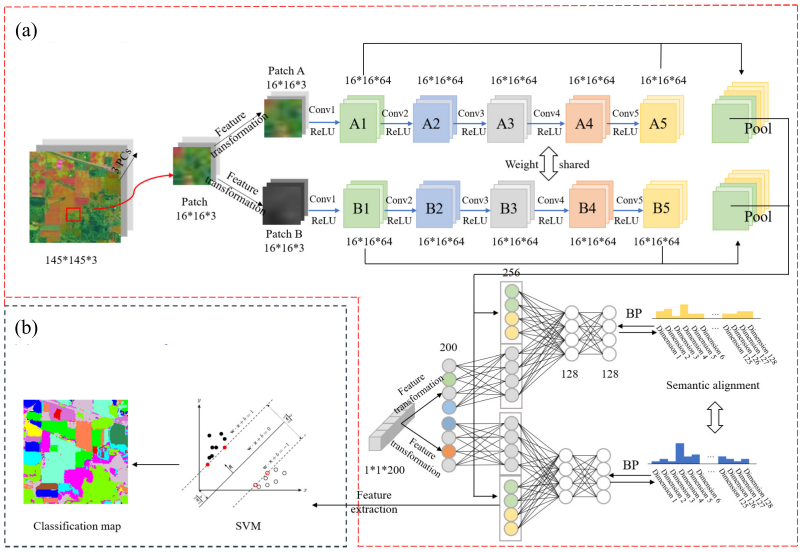


Fig. 7. (a) Framework of the proposed S3FN. The S3FN utilizes Spe-SFN and Spa-SFN to extract the spectral and spatial features, respectively, and then adaptively merges them followed by a merging layer. (b) Merged features are processed by a classifier.

where M_{i-1} denotes the feature map in the $(i-1)$th layer, P_i and Q_i are the height and the width of the convolutional layer, w_{ijm}^{pq} indicates the weight of the 3×3 convolutional kernel at position (p, q), which is connected to the previous mth feature map, u_{ij}^{xy} represents the value at position (x, y) of the jth feature map in the ith layer, and b_{ij} is the bias for the jth feature map of the ith layer. Thus, the area of the receptive field can grow with the increase of the number of convolutional layers. According to [56], the 3×3 convolutional kernel avoids expression bottlenecks and enhances the nonlinear expression ability.

Features in the deep layers are more abstract and robust than those in the shallow layers. Information loss, however, is a risk as the convolutional layer gets deeper. Therefore, in the proposed spatial feature learning network, we merge the first and fifth convolutional layers in the merging layer with a concatenation operation followed by a global average pooling layer. The global average pooling was developed to enforce correspondences between feature maps and categories, which can be interpreted as category confidence maps [57]. By taking the average of each feature map, the global average pooling layer is robust to spatial translations of the input. As shown in Fig. 6, the input is a $6 \times 6 \times 3$ 3-D feature map, and after global average pooling, this feature map is transformed into output $1 \times 1 \times 3$, that is, the values in each layer will be averaged into one value. The corresponding pooling feature can be calculated as

$$h = D_{\text{avg}}(g(U_1 \circ U_5)) \tag{10}$$

where U_i represents the convolutional feature map of ith layer, the $\circ$ symbol represents a concatenation operation and D_{avg} is the global average pooling. Finally, the merge layer is connected to two fully connected layers.

The multiview feature descriptions of the same source pixels are projected into deep embedding space by the spatial feature learning network. The embedding features can be formulated as

$$c_{\text{spa}} = f_{\text{spa}}(\phi_{\text{spa}}(v_{\text{spa}})) \tag{11}$$

$$c_{\text{spa}}^0 = f_{\text{spa}}\left(\phi_{\text{spa}}\left(v_{\text{spa}}^0\right)\right) \tag{12}$$

where ϕ_{spa} represents the network with five 2-D convolutional layers and a global average pooling layer, f_{spa} denotes the last two fully connected layers.

Similar to Spe-SFN, the high-level semantic alignment strategy in Spa-SFN requires consistent spatial representation between multiple views of the transformed pixels as follows:

$$\mathcal{L}_{\text{spa}} = -\sum_{i \in I} log \frac{\exp\left(c_{\text{spa}(i)} \cdot c_{\text{spa}(i)}^0 / \tau\right)}{\sum_{a \in Ai} \exp\left(c_{\text{spa}(a)} \cup c_{\text{spa}(I)}^0\right) / \tau)}. \tag{13}$$

After training, the feature learning network is regarded as a feature extractor. The output of the pooling layer is extracted as a spatial feature since shallow layers should have more detailed spatial information. As shown in Fig. 2(c), features extracted by Spa-SFN are more abstract than those extracted by reconstruction-based FE method 3-D-CAE because of the high-level semantic information considered in our framework.

D. Spectral–Spatial Semantic Feature Learning for HSI Classification

The integration of spectral and spatial features is vital for HSI classification. In this section, we integrate the spectral and spatial FE networks into a unified framework, as shown in Fig. 7. First, we obtain feature descriptions of spectral signal with (1) and (2) and the spatial signal with (7) and (8). Subsequently, the spectral–spatial feature learning network was designed with spectral and spatial feature extractors in Spe-SFN and Spa-SFN, respectively, to extract hierarchical spectral–spatial features. Then, the two extractors adaptively merge in the merging layer, which can be formulated as

$$h_{\text{ss}} = \lambda_{\text{spe}}\phi_{\text{spe}}(v_{\text{spe}}) \circ \lambda_{\text{spa}}\phi_{\text{spa}}(v_{\text{spa}}) \tag{14}$$

$$h_{\text{ss}}^0 = \lambda_{\text{spe}}\phi_{\text{spe}}\left(v_{\text{spe}}^0\right) \circ \lambda_{\text{spa}}\phi_{\text{spa}}\left(v_{\text{spa}}^0\right) \tag{15}$$

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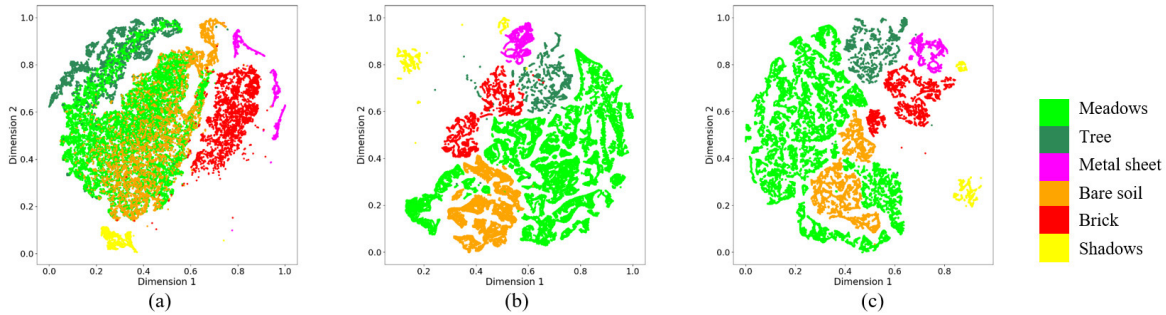


Fig. 8. Data distribution for the six categories of the reference samples from the Pavia University dataset in the learned feature space of (a) Spe-SFN, (b) Spa-SFN, and (c) S3FN.

Algorithm 1 S3FN

Input: HSI X, parameter τ, training epoch E.

1: Crop the spectral signal $x_{spe} \in R^n$ and the spatial signal $x_{spa} \in R^{w \times w \times p}$.
2: Obtain multi-view spectral features v_{spe}, v_{spe}^0 via (1) and (2), and spatial features v_{spa}, v_{spa}^0 via (7) and (8).
3: Initialize weights of the network with random values, and input multi-view feature descriptions into the S3FN.
4: **for** n $= 1, 2, \ldots, E$ **do**
5: Obtain hierarchical spectral–spatial feature and merge both parts with (14) and (15).
6: Obtain spectral–spatial feature in the deep embedding space with (16) and (17).
7: Calculate semantic loss with (18), and optimize the network with the stochastic gradient descent algorithm.
8: **end for**

Output: The unsupervised learned features for each pixel.

where the $\circ$ symbol represents a concatenation operation, λ_{spe} and λ_{spa} are adaptive weighting factors, which are trained together with the other parameters in the network. Finally, the merging layer is connected to two fully connected layers, called f_{ss}. The spectral–spatial feature representation in the embedding space can be calculated as

$$c_{\text{ss}} = f_{\text{ss}}(h_{\text{ss}}) \tag{16}$$

$$c_{\text{ss}}^0 = f_{\text{ss}}\left(h_{\text{ss}}^0\right). \tag{17}$$

Therefore, the training loss in S3FN can be defined as

$$\mathcal{L}_{\text{ss}} = -\sum_{i \in I} \log \frac{\exp\left(c_{\text{ss}(i)} \cdot c_{\text{ss}(i)}^0 / \tau\right)}{\sum_{a \in Ai} \exp\left(c_{\text{ss}(a)} \cup c_{\text{ss}(I)}^0\right) / \tau)}. \tag{18}$$

The implementation steps of the S3FN are summarized in Algorithm 1.

To further investigate the learned features of related spectral, spatial and spectral–spatial methods, we visualize the data distribution for the six categories of the reference samples from the Pavia University dataset in Fig. 8. We apply PCA to project the data from its original high-dimensional space onto a 2-D space for visualization. It can be seen that in the learned space, the separation distance between different categories of Spe-SFN in Fig. 8(a) and Spa-SFN in Fig. 8(b), is relatively smaller than that of S3FN in Fig. 8(c), which demonstrates the advantage of combining spectral and spatial information.

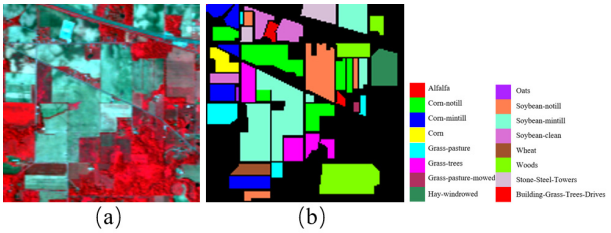


Fig. 9. Indian Pines dataset. (a) Three-band false-color composite. (b) Ground-truth map.

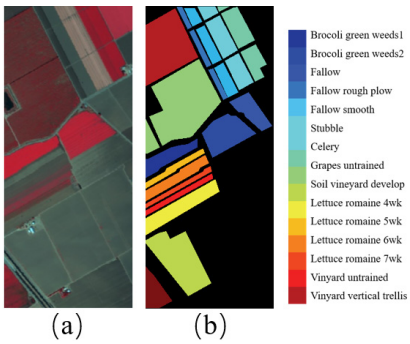


Fig. 10. Salinas dataset. (a) Three-band false-color composite. (b) Ground-truth map.

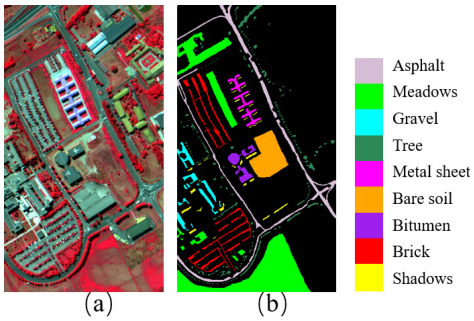


Fig. 11. Pavia University dataset. (a) Three-band false-color composite. (b) Ground-truth map.

IV. Experiments

A. Dataset Description

The proposed S3FN was validated on three benchmark HSI datasets to illustrate its effectiveness.

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE INDIAN PINES DATASET

Class number	Class name	Training	Test
1	Alfalfa	7	39
2	Corn-notill	215	1213
3	Corn-minitill	125	705
4	Corn	36	201
5	Grass-pasture	73	410
6	Grass-trees	110	620
7	Grass-pasture-mowed	5	23
8	Hay-windrowed	72	406
9	Oats	3	17
10	Soybean-notill	146	826
11	Soybean-minitill	369	2086
12	Soybean-clean	89	504
13	Wheat	31	174
14	Woods	190	1075
15	Bldg-Grass-Trees-Drives	58	328
16	Stone-Steel-Towers	14	79
	Total	1543	8706

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE SALINAS DATASET

Class number	Class name	Training	Test
1	Broccoli green weeds 1	100	1909
2	Broccoli green weeds 2	186	3540
3	Fallow	99	1877
4	Fallow rough plow	70	1324
5	Fallow smooth	134	2544
6	Stubble	198	3761
7	Celery	179	3400
8	Grapes untrained	564	10707
9	Soil vineyard develop	310	5893
10	Corn senesced greed weeds	164	3114
11	Lettuce romaine, 4 wk	53	1015
12	Lettuce romaine, 5 wk	96	1831
13	Lettuce romaine, 6wk	46	870
14	Lettuce romaine, 7 wk	54	1016
15	Vineyard untrained	363	6903
16	Vineyard vertical trellis	90	1717
	Total	2706	51421

1) *Indian Pines:* The dataset covers the Indian Pines test site in Northwestern Indiana, U.S.A., and was collected by the Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. The size of this dataset is $145 \times 145 \times 220$, however, $145 \times 145 \times 200$ were retained in our experiments because of the removal of 20 water absorption bands. In summary, this dataset has 16 classes and contained 10 249 labeled pixels. Fig. 9 shows the false-color composite of the Indian Pines in Fig. 9(a) and the ground reference map in Fig. 9(b). Table I lists the numbers of the training and test samples.
2) *Salinas:* This dataset covers the the Salinas Valley, California, and was acquired by the AVIRIS sensor at a high spatial resolution of 3.7 m. Salinas is of a spatial size of 512×217 pixels and 224 spectral reflectance bands ranging from 400 to 2500 nm. After removing noisy and water absorption bands, 204 bands were used for the analysis. The false-color image and the ground-truth map are shown in Fig. 10. Table II lists the training and test samples per class.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR THE UNIVERSITY OF PAVIA DATASET

Class number	Class name	Training	Test
1	Asphalt	331	6300
2	Meadows	932	17717
3	Gravel	104	1995
4	Trees	153	2911
5	Metal sheets	67	1278
6	Bare soil	251	4778
7	Bitumen	66	1264
8	Bricks	184	3498
9	Shadows	47	900
	Total	2135	40641

3) *University of Pavia:* This scene covers the University of Pavia, Italy, and was gathered by the reflective optics system imaging spectrometer (ROSIS) sensor. This dataset includes 610×340 pixels and 103 reflectance bands after removing the water absorption bands. The University of Pavia ground truth includes 42 776 labeled samples from nine classes. The false-color image and ground-truth map can be seen in Fig. 11. Table III lists the training and test sets.

B. Comparative Methods and Experimental Setup

The comparative and proposed methods and their implementation details are described as follows.

1) *SVM [50]:* Spectral classification with the radial basis function (RBF) SVM, which was implemented using the library for SVM (LIBSVM). The parameters C and σ are from 2^{-9} to 2^{7} for fivefold cross-validation.
2) *PCA:* The first 30 principal components were classified by the RBF-SVM. The grid search ranging from 2^{-9} to 2^{7} was applied to obtain the optimal parameters C and σ.
3) *SAE_LR [42]:* Spectral–spatial classification with SAE. We followed the implementation in [42].
4) *CNN [47]:* Spatial classification with $16 \times 16 \times 1$ input patch via a 2-D-CNN. We followed the implementation in [47].
5) *3-D-CAE [43]:* 3-D unsupervised spatial–spectral feature learning method, with 3-D convolutional kernels to improve unsupervised feature learning performance. We followed the implementation in [43].
6) *3-D-SACNN [44]:* Unsupervised spectral–spatial feature learning with a 3-D encoder–decoder architecture. We followed the implementation in [44].
7) *Spe-SFN:* The proposed spectral SFN, which was trained using Adam optimizer with a learning rate of 1e-4 and a weight decay of 1e-6. The detailed descriptions of the framework are presented in Section III-B.
8) *Spa-SFN:* The proposed spatial SFN. The detailed descriptions of the framework are presented in Section III-C.
9) *S3FN:* The proposed spectral–spatial SFN. The detailed descriptions of the framework are presented in Section III-D.

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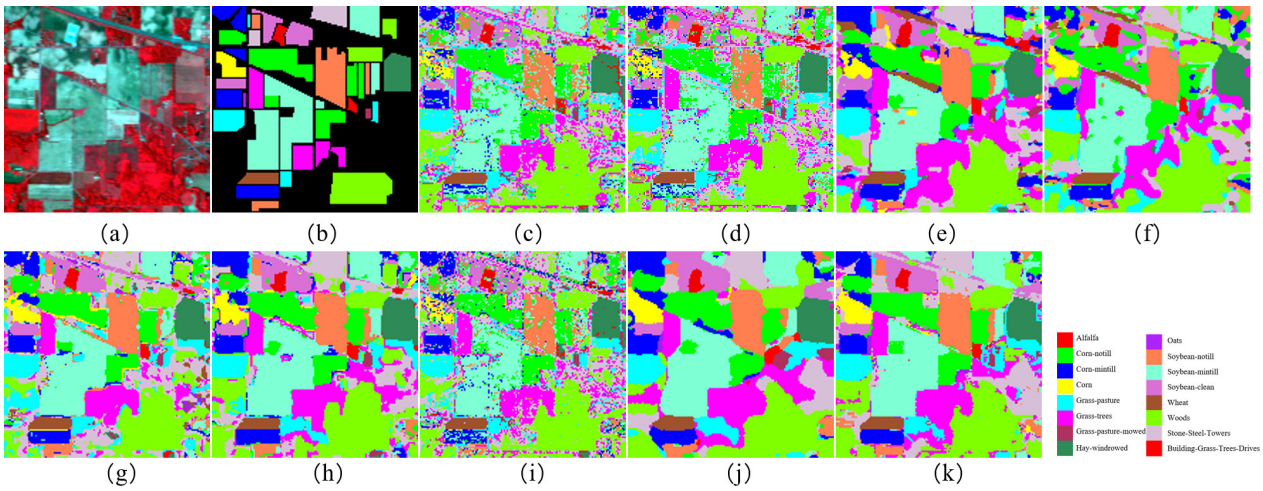


Fig. 12. Classification maps for the Indian Pines dataset. (a) False-color image. (b) Ground-truth map. (c) Raw. (d) PCA. (e) SAE_LR. (f) CNN. (g) 3-D-CAE. (h) 3-D-SACNN. (i) Spe-SFN. (j) Spa-SFN. (k) S3FN.

TABLE IV
CLASSIFICATION RESULTS OF FEATURE LEARNING METHODS WITH 15% TRAINING SAMPLES ON THE INDIAN PINES DATASET

Class	Raw	PCA	SAE_LR	CNN	3-D-CAE	3-D-SACNN	Spe-SFN	Spa-SFN	S3FN
1	31.71	49.74	85.38	72.05	**100.00**	**100.00**	48.72	90.00	94.62
2	75.95	75.94	92.09	80.33	94.91	94.89	77.97	**97.97**	97.96
3	61.31	67.30	95.55	77.32	90.80	97.87	69.01	98.31	**99.18**
4	49.77	62.79	83.43	89.85	95.06	89.05	62.79	98.36	**98.31**
5	92.64	89.41	94.37	87.34	97.23	**98.54**	88.02	96.54	97.95
6	97.26	95.00	97.05	94.26	99.80	97.58	94.87	98.74	**99.81**
7	84.00	82.17	91.74	78.26	97.06	**100.00**	86.96	98.26	98.70
8	97.91	97.34	97.93	98.35	**100.00**	99.75	97.17	99.95	99.98
9	38.89	53.53	73.53	77.65	**98.70**	94.12	46.47	88.24	97.06
10	73.94	74.53	92.72	81.14	91.07	94.07	74.59	97.37	**98.21**
11	83.75	82.20	97.25	94.15	95.30	98.47	81.68	98.73	**99.29**
12	75.84	65.73	89.70	69.38	94.30	92.86	70.36	96.92	**97.62**
13	98.91	96.32	98.74	97.82	97.63	**100.00**	96.44	98.68	99.20
14	96.57	94.41	98.37	97.60	99.41	98.60	94.66	99.40	**99.91**
15	57.06	54.24	91.22	83.38	91.22	97.56	56.01	97.99	**99.09**
16	89.29	88.48	92.78	**98.73**	96.31	84.81	92.15	93.29	98.48
OA (%)	81.33	80.54	94.87	87.94	94.71	96.85	81.16	98.25	**98.91**
AA (%)	75.30	77.74	94.15	86.10	96.18	96.14	77.37	96.80	**98.46**
Kappa × 100	78.62	76.82	91.99	86.19	94.05	96.41	78.47	98.00	**98.76**

The deep learning-based methods were conducted in PyTorch. The training was powered by an NVIDIA Tesla V100 32 GB GPU. The other compared methods were implemented with MATLAB 2014a. The performances of state-of-the-art (SOTA) deep learning-based unsupervised feature learning methods were compared to illustrate the effectiveness of the proposed frameworks. We measured the performance of the methods with the overall accuracy (OA), average accuracy (AA), and Kappa coefficient. To make a fair comparison, we repeated all the experiments ten times with random training and test sets. Intuitively, the higher the accuracy, the stronger the feature learning ability of the method.

C. Classification Results

The classification maps of the compared methods are shown in Figs. 12–14. The results show that due to the lack of spatial information, spectral feature classification methods like Raw, PCA, and Spe-SFN usually suffer from salt and pepper noise, indicating that the continuity cannot be maintained within the targets. In comparison, spatial feature classification methods including CNN and the proposed Spa-SFN achieve smoother results due to the consideration of neighborhood information. However, conventional CNN cannot obtain satisfactory classification results when labeled samples are limited. We also observed that although Spa-SFN enables the network to produce smoother results, the accuracy in the boundary is sacrificed. As shown in Fig. 14(j), the over-smoothing phenomena or even misclassified results are apparent, since there are many isolated small targets in the Pavia University dataset. Considering the characteristics of spectral and spatial features, we intuitively integrated them to yield a more accurate classification result. It can be seen that the spectral–spatial-based SAE_LR also fails to obtain satisfactory

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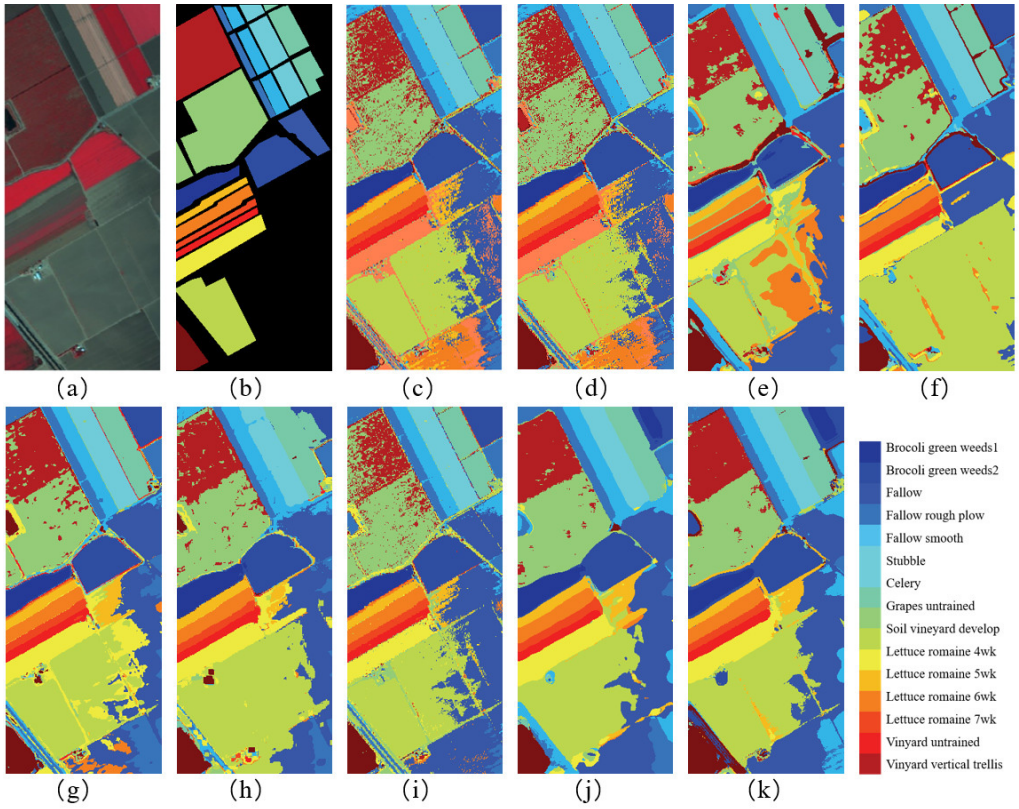


Fig. 13. Classification maps for the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) Raw. (d) PCA. (e) SAE_LR. (f) CNN. (g) 3-D-CAE. (h) 3-D-SACNN. (i) Spe-SFN. (j) Spa-SFN. (k) S3FN.

TABLE V
CLASSIFICATION RESULTS OF FEATURE LEARNING METHODS WITH 5% TRAINING SAMPLES ON THE SALINAS DATASET

Class	Raw	PCA	SAE_LR	CNN	3-D-CAE	3-D-SACNN	Spe-SFN	Spa-SFN	S3FN
1	99.11	99.15	88.19	99.72	99.31	99.31	99.23	92.30	**99.95**
2	99.36	**99.75**	95.98	97.86	99.55	97.35	99.49	98.00	99.60
3	99.12	99.20	96.28	92.76	99.47	99.48	99.25	98.51	**100.00**
4	99.03	99.03	98.47	98.23	98.50	93.75	99.03	99.15	**99.40**
5	98.44	98.36	99.45	98.59	97.98	97.98	98.20	96.85	**99.69**
6	99.85	99.85	99.55	99.93	99.78	**100.00**	99.74	98.70	**100.00**
7	99.63	99.66	97.81	99.00	99.40	99.40	99.49	99.13	**99.94**
8	88.32	87.88	88.70	89.53	93.37	93.37	88.38	**99.95**	98.22
9	99.79	99.72	96.97	99.31	99.78	99.78	99.56	88.23	**99.95**
10	95.43	95.16	95.48	89.85	96.69	96.69	94.67	97.55	**98.91**
11	97.91	97.76	94.65	91.34	97.72	97.72	96.48	98.97	**100.00**
12	99.16	99.07	98.27	99.34	99.97	99.97	99.14	97.04	**100.00**
13	98.44	98.61	96.47	99.34	98.98	97.63	96.70	98.67	**100.00**
14	95.84	95.91	98.55	97.87	97.50	**99.41**	94.17	99.40	99.31
15	70.88	69.36	90.66	75.47	90.00	91.22	66.06	**97.95**	97.44
16	98.64	98.61	97.38	91.21	97.78	96.31	98.24	92.53	**100.00**
OA (%)	92.93	92.59	94.33	92.68	96.60	97.58	92.06	98.39	**99.13**
AA (%)	92.12	91.74	93.70	94.96	97.87	97.98	95.49	97.06	**99.52**
Kappa × 100	96.24	96.07	95.80	91.84	96.21	97.01	91.14	98.16	**99.04**

classification results when the number of labeled samples is limited. Deep learning-based unsupervised spectral–spatial feature learning methods, including 3-D-CAE, 3-D-SACNN, and the proposed S3FN outperformed the other comparative methods. However, since 3-D-CAE and 3-D-SACNN pay more attention to the low-level features, the classification maps are still affected by salt and pepper noise, as shown in Figs. 12–14. By inheriting the superiority from Spe-SFN and Spa-SFN, the proposed S3FN makes a tradeoff between the target continuity and the boundary information. As shown in Figs. 12(k), 13(k), and 14(k), S3FN maintains the continuity inside targets and also obtains clear boundaries of the targets.

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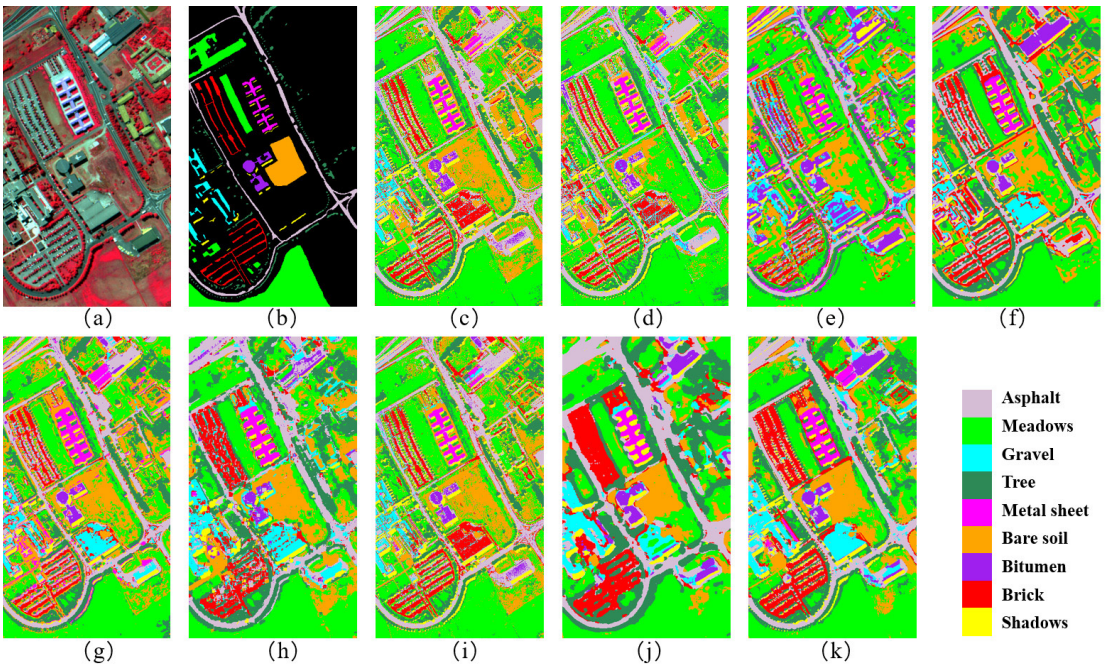


Fig. 14. Classification maps for the Pavia University dataset. (a) False-color image. (b) Ground-truth map. (c) Raw. (d) PCA. (e) SAE_LR. (f) CNN. (g) 3-D-CAE. (h) 3-D-SACNN. (i) Spe-SFN. (j) Spa-SFN. (k) S3FN.

TABLE VI
CLASSIFICATION RESULTS OF FEATURE LEARNING METHODS WITH 5% TRAINING SAMPLES ON THE PAVIA UNIVERSITY DATASET

Class	Raw	PCA	SAE_LR	CNN	3-D-CAE	3-D-SACNN	Spe-SFN	Spa-SFN	S3FN
1	92.62	92.69	89.20	96.16	96.01	94.32	91.63	98.42	**98.43**
2	97.81	97.89	96.54	98.93	98.82	98.52	97.81	99.31	**99.55**
3	77.20	77.18	66.35	66.36	88.71	69.17	79.95	95.91	**96.18**
4	92.43	92.34	95.85	97.37	94.70	95.95	93.30	96.23	**98.86**
5	99.05	98.86	96.85	**100.00**	99.62	98.36	99.37	99.50	99.94
6	87.75	87.44	64.42	84.62	89.33	94.01	86.77	**98.70**	98.14
7	84.88	84.72	77.98	81.68	89.94	80.38	83.47	**94.51**	95.62
8	88.91	89.24	84.70	89.52	92.09	87.08	89.65	97.40	**98.88**
9	99.80	99.77	96.61	99.97	99.04	88.33	**100.00**	91.19	98.87
OA (%)	93.34	93.36	88.51	93.82	95.65	94.04	93.30	98.22	**98.81**
AA (%)	91.14	91.16	84.57	90.51	94.25	89.57	91.33	96.80	**98.27**
Kappa × 100	91.16	91.12	85.39	91.75	94.21	92.32	91.09	97.64	**98.42**

The quantitative assessments of different methods are shown from Tables IV–VI. It can be observed that the quantitative results are consistent with visual analysis. The spatial feature classification methods generally outperform the spectral feature classification methods including Raw, PCA, and Spe-SFN. Compared with conventional CNN and SAE_LR, deep learning-based unsupervised feature learning methods, such as 3-D-CAE, 3-D-SACNN, and the proposed S3FN, have more powerful classification capabilities with fewer training samples. Benefiting from the high-level spectral–spatial semantic feature learning, the proposed S3FN delivered robust performances with an OA higher than 98% for three datasets, which outperformed other compared methods. The S3FN method performed well on both the small datasets (Indian Pines) and the large datasets (Salinas and Pavia University), which demonstrate the robustness of learned spectral–spatial features when dealing with different datasets.

TABLE VII
RUN TIME COMPARISON BETWEEN UNSUPERVISED FE METHODS BASED ON DEEP LEARNING

Dataset	Run Time (s)				
	3-D-CAE	3-D-SACNN	Spe-SFN	Spa-SFN	S3FN
Indian Pines	939.08	1448.30	304.34	314.07	344.12
Salinas	4399.38	6001.75	1664.20	1790.50	1797.60
Pavia University	3538.90	5594.30	1429.70	1596.80	1618.80

D. Parameters Analysis

1) Time Cost: The run time is also a factor that restricts the development of deep learning-based methods in the remote-sensing field. Therefore, we assess the mean time cost of ten repeat times of the proposed Spe-SFN, Spa-SFN, and S3FN along with the comparative deep learning-based unsupervised feature learning methods. From Table VII, it can be concluded

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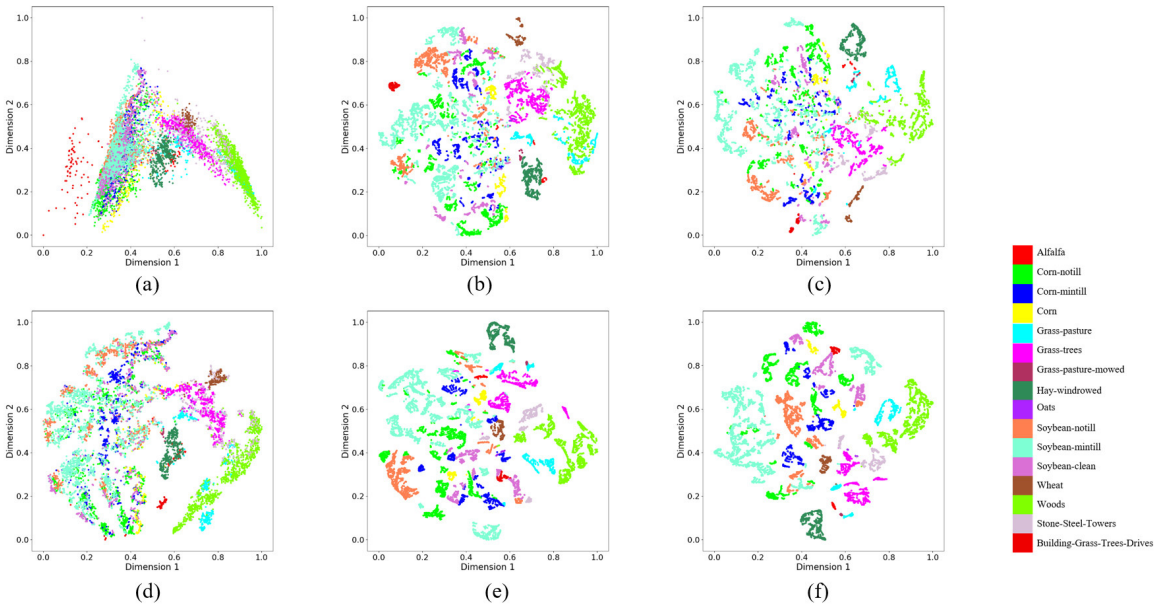


Fig. 15. Data distributions of the reference samples of the Indian Pines dataset in the original spectral feature space (a) and the learned feature space (b) 3-D-CAE, (c) 3-D-SACNN, (d) Spe-SFN, (e) Spa-SFN, (f) S3FN, respectively. PCA was applied to map the data from original high-dimensional space into a 2-D space for visualization.

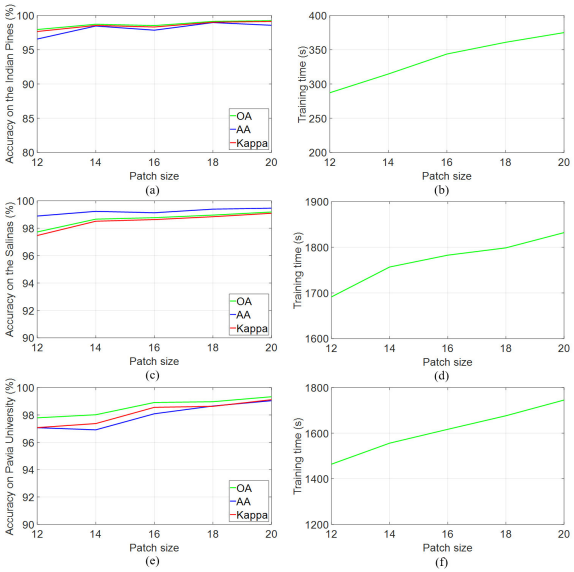


Fig. 16. Classification accuracy (the first column) and execution time (the second column) under different spatial size of input data, respectively. (a) and (b) Indian Pines dataset. (c) and (d) Salinas dataset. (e) and (f) Pavia University dataset.

the time cost of the proposed S3FN is much less than the 3-D convolution kernels-based methods for a variety of datasets. This is mainly because of the proposed efficient feature learning network to learn spectral and spatial features separately. Taking Salinas dataset as an example, the run time of the proposed S3FN is about 1797 s while that of 3-D-CAE is 4399 s and that of 3-D-SACNN is 6001 s. We observed that Spe-SFN and Spa-SFN cost less time than S3FN; however, their accuracy can hardly be guaranteed since only spectral or spatial characteristics are considered. Therefore, we conclude that the proposed S3FN can achieve an optimal balance between accuracy and efficiency.

2) Feature Analyses: In this section, we visualize the data distributions of the reference samples of the Indian Pines dataset in the learned feature space to investigate the separability performance of FE methods. The reference samples of the Indian Pines dataset was mapped into a 2-D space using PCA for visualization. The results are shown in Fig. 15. As we can see in Fig. 15(a), there is limited separability between different categories in the original feature space. In the feature spaces of 3-D-CAE and 3-D-SACNN, the data distribution of different classes is relatively separable. However, the distribution of different classes still presents some mixed-up phenomena due to the limitations of low-level features. We also find that the distribution of Spa-SFN is more separable than that of Spe-SFN, which further demonstrates the contribution of spatial information to FE. The data distribution of different classes in the feature space of S3FN is most separable. Meanwhile, pixels within the same class lie closer together in the feature space.

3) Influence of Patch Size: The neighborhood size of input data influences classification accuracy and efficiency. We set different patch sizes: 12×12, 14×14, 16×16, 18×18, and 20×20 for the input data. The results are shown in Fig. 16. The columns on the left of Fig. 16 correspond to the classification accuracy and the columns on the right side are the run times of the S3FN. It can be observed that an increase in patch size can enrich spatial information, which first improves the classification performance, and then goes stable. However, the redundant computation inevitably occurs because of the large overlaps among adjacent patches, so the time cost of the S3FN increases with an increase in the patch size. In addition, as the patch size increases, the accuracy of isolated small targets and boundaries will be sacrificed, resulting in an over-smoothing phenomenon. Therefore, we chose the intermediate value 16×16 as the patch size in our method, achieving a balance between classification performance and run time.

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V. Conclusion

In this article, we propose a novel end-to-end S3FN. First, we obtain the feature descriptions of the same source data under multiviews via feature transformation. Subsequently, we propose the learnable spectral–spatial feature learning network to project the two feature descriptions into the deep embedding feature space. Then, a contrastive loss is introduced to complete the high-level semantic alignment strategy. To improve the efficiency of the method, we designed efficient networks to learn spectral and spatial features separately and combined both parts to execute the subsequent contrastive learning. Unlike data reconstruction-based FE methods, the proposed S3FN can learn high-level semantic representation from data, avoid latching on the low-level features, and therefore increase the performance of subsequent semantic classification task. Experimental results show that the proposed S3FN yields competitive classification results with lower time cost compared to other unsupervised deep learning-based FE methods on various datasets.

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